PVG’s College of Engineering and Technology and GKPIOM, Pune.

Department of Information Technology.

B.E. IT PROJECT

[Academic Year 2025-26]

Date: 03/09/25

SYNOPOSIS

PROJECT TITLE: OIL SLICK DETECTION and DRIFT PREDICTION USING SATELLITE IMAGERY

ABSTRACT

The monitoring of oil slicks is essential for safeguarding marine ecosystems, averting environmental catastrophes, and upholding maritime regulations. However , existing detection models face significant limitations ,as SAR dark spot analysis often generates false positives from natural oceanic features like low wind zones and biogenic slicks, traditional approaches rely on intensity values with limited semantic and textural feature analysis, detection depends upon manual interpretation making it time consuming and difficult to scale. In addition, ancillary data like wind speed, currents, sea surface conditions are often under utilized. To address these limitations, we propose to develop an automated oil slick detection framework that integrates high-resolution Synthetic Aperture Radar (SAR) imagery with ancillary data sources. Our approach will combine SAR dark spot detection algorithms with semantic and textural analysis techniques, integrated through supervised deep learning methods for detection modeling. Our pipeline will be designed to distinguish oil slicks from dark oceanic features through comprehensive feature analysis and classification. We plan to establish a robust, scalable framework that enhances maritime environmental monitoring capabilities.

Keywords: Oil slick detection, satellite imagery, SAR, dark spot detection, deep learning algorithms, environmental data, oil pollution, remote sensing, maritime monitoring, ocean surface analysis, proactive detection, pollution prevention, coastal resilience.

INTRODUCTION

Oil slicks, characterized as thin layers of oil floating on the ocean surface pose a serious threat to marine ecosystems, coastal environments, and maritime industries. They originate from a variety of sources, including shipping operations, accidental spills, natural hydrocarbon seeps, and offshore drilling. Even small-scale, chronic slicks can have cumulative ecological impacts, affecting water quality, marine flora and fauna, and local economies dependent on fishing and tourism.

Traditional monitoring of oil slicks relies on ship-based patrols, aerial surveillance, and manual image interpretation, which are often expensive, spatially limited, and temporally infrequent**.** These methods fall short in providing the continuous, large-scale surveillance required to detect early-stage slicks or monitor recurring events like natural seepage.

In contrast, **satellite-based Imagery Earth Observation** using **Synthetic Aperture Radar (SAR)** sensors on platforms like **Sentinel-1, Sentinel-2** - has emerged as a powerful alternative. SAR can detect oil slicks regardless of cloud cover or sunlight, making it ideal for oil slick monitoring over vast oceanic regions. When combined with **multi-temporal analysis**, **environmental variables** (e.g., wind, currents), and **deep learning models**, satellite data can be used **not only to detect** but also to **predict the movement of oil slicks**. The pipeline incorporates **dark spot detection**, **look-alike discrimination**, **spatio-temporal feature extraction** to analyze oil slicks. Emphasis is placed on evaluating how environmental conditions influence their persistence and spread.

This approach enables scalable, high-frequency monitoring essential for sustainable ocean governance in the face of increasing offshore activities and climate variability.

LITERATURE SURVEY

Sr. No	Name and Year of Publish	Work description	Problems found
1.	Oil Slick Segmentation Using Deep Learning: A Framework for Real-World Applications Based on Sentinel-1 Imagery. [1] Published - 2025	- Deep learning based approach for global oil slick - Segmentation using Sentinel-1 SAR imagery. - Proposing a “snowballing framework” improves data quality and diversity.	- Under represented lookalike samples in some regions - Model parameter limit: billions of parameters - No direct comparison with many previous state-of-the-art methods.
2.	Detection of Oil Slicks in SAR Satellite Images Using Otsu-Bradley’s Thresholding Method. [2] Published - 2025	- Proposed a novel thresholding method combining Otsu’s global thresholding and Bradley’s local thresholding for improved oil slick segmentation in SAR imagery. - Tested on Sentinel-1 datasets, achieving higher detection accuracy and robustness against noise.	- Performance decreases in complex sea states with high clutter; requires parameter tuning for different environmental conditions. - Works well only on smaller and areas having detailed features.

3.	Oil Spill Detection in PolSAR Imagery Using Composite Scattering Power Entropy and Multi-Scale Hybrid Feature Fusion Network. [3] Published in 2025	- Proposed Hc feature combining Cloude entropy (H) and Freeman power entropy (Hp) for better discrimination. - Developed MSHFFNet integrating Hc with polarimetric intensity for segmentation. - Used hybrid pooling (square + strip) for elongated slick detection. - Applied feature fusion with sSE blocks for better small-scale detection.	- - Difficulty distinguishing oil slicks from look-alikes - - Poor detection accuracy for small-scale oil slicks. - - Missed or incomplete detection of elongated oil slicks. - - Low boundary precision between oil and seawater. - Interference from sea clutter and complex marine backgrounds.
4.	Oil Slick Identification in Marine Radar Image using HOG, Random Forest and PSO. [4] Published - 2024	- Proposed an oil spill detection method combining Histogram of Oriented Gradients (HOG) feature extraction, Random Forest classification, and Particle Swarm Optimization (PSO) for adaptive dual-threshold segmentation.	- Some false positives remain in ship wake regions; effectiveness decreases in extreme sea states (too calm or too rough). - Involves manual intervention for some processing steps.

		- Applied preprocessing steps like interference suppression, speckle removal, and contrast enhancement on X-band marine radar images. - Achieved accurate segmentation of real oil slicks and reduced false positives.	
5.	Performance Measures for Validation of Oil Spill Dispersion Models Based on Satellite and Coastal Data. [5] Published - 2024	- Introduced new performance measures, Area Overlap Measure (AOM) and Oil Mass Capture (OMC), for validating oil spill dispersion models. - Used Sentinel-1 and Sentinel-2 satellite data to generate ground-truth datasets for model validation. - Demonstrated the effectiveness of these measures through a simulated oil spill case study.	- Existing validation methods are often qualitative or rely on single metrics, which can be insufficient. - Difficulty of obtaining high-quality ground-truth data from satellite imagery due to cloud cover and atmospheric conditions. - Using a **simulated spill** means challenges of real-world, large-scale applications remain unaddressed.

6.	Solution for Modelling the Marine Oil Spill Drift (Article). Published - 2025	- Provided a comprehensive review of remote sensing techniques for oil spill detection and monitoring. - Discussed various sensor types, including Synthetic Aperture Radar (SAR), optical, and thermal infrared sensors. - Explained the physical principles behind how each sensor detects oil spills.	- Difficulty in distinguishing oil slicks from “look-alikes” such as calm water. - Optical sensors are constrained by cloud cover and lack of daylight. - Environmental factors such as wind speed and sea state can affect detection accuracy.

PROPOSED WORK OF PROJECT

This project aims to develop an **automated oil slick detection and drift prediction system** by integrating high-resolution Synthetic Aperture Radar (SAR) satellite imagery with environmental datasets. The system will identify oil slicks by detecting “dark spots” in radar images—common indicators of oil while using texture and shape analysis to distinguish them from look-alikes such as calm water patches or natural films. Deep learning models, particularly convolutional neural networks (CNNs), will be trained to enhance classification accuracy.

The initial phase will focus on building a **slick detection pipeline** capable of processing SAR images using both conventional dark spot detection techniques for identifying slick regions and CNN-based feature classification to confirm whether the detected dark spot is an oil slick . Multiple detection models and methodologies will be evaluated for their strengths, limitations, and suitability for environmental monitoring. The most effective approach will be refined into a robust framework suitable for long-term, large-scale analysis of marine pollution trends. The framework will be designed for modularity, ensuring it can adapt to improvements in SAR technology, higher-resolution sensors, and future AI models without major redesign.

Following the detection phase, the framework will incorporate a **drift prediction module** to estimate oil slick movement over time. This module will integrate real-time oceanographic and meteorological data, including surface current velocity fields, wind speed and direction, tidal patterns, and wave conditions. Surface currents will be the primary driver for large-scale displacement, while wind-induced drift estimated at approximately 3% of wind speed will model downwind movement. Wave-induced Stokes drift will capture near-surface transport effects, and tidal oscillations will account for complex coastal dynamics.

The drift model will be dynamically re-forced with newly available environmental data ensuring that predictions remain adaptive to evolving ocean dynamics. By continuously integrating this information, the model improves forecast accuracy from the moment of detection until containment. Integrating detection with drift modeling enables proactive response strategies, supporting faster and more targeted containment operations while minimizing environmental damage.

In the long term, the system will serve as a **scalable monitoring platform** for tracking marine pollution patterns. Its outputs will be valuable for environmental agencies, policymakers, and disaster response teams, offering data-driven insights to improve preparedness, guide protective measures, and reduce the ecological footprint of oil spills. Beyond immediate response applications, the accumulated detection and drift data will contribute to research on spill behavior, environmental resilience, and long-term ocean health monitoring.

REQUIREMENTS

A) HARDWARE REQUIREMENTS:

• High-performance GPU (NVIDIA RTX 4060): To handle the computational complexity of training deep learning models, which require large amounts of memory and processing power.

• At least 8GB of RAM: For efficient handling of large satellite image datasets.

• Storage (SSD): For storing the Satellite image datasets (in PNG or tiff) and intermediate training outputs.

• Multi-core CPU: To support parallel processing during model training and inference.

B) SOFTWARE REQUIREMENTS:

• Operating System: Linux (Ubuntu preferred) for compatibility with deep learning libraries and GPU acceleration.

• Python 3.8+: Programming language for writing the deep learning model and preprocessing code.

• TensorFlow or PyTorch: Deep learning frameworks for building and training the model.

• CUDA and cuDNN: NVIDIA libraries to enable GPU acceleration.

• Satellite Images: API for collecting live satellite images.

• Weather Data: For predicting the drift direction of oil slicks.

• Visualization Tools (e.g., Matplotlib, Seaborn): For visualizing training results, loss curves, and generated outputs.

EXPECTED RESULT

The project is expected to produce a **scalable, high-accuracy oil spill monitoring framework** that integrates advanced SAR image processing, deep learning classification, and drift prediction. Optimal preprocessing steps—such as speckle noise reduction, radiometric calibration, and feature extraction—will improve image clarity and detection reliability. Combining classical dark spot detection with texture and semantic feature analysis will increase robustness against look-alikes and environmental variability.

Supervised deep learning models, particularly CNNs, will enhance the distinction between genuine oil slicks and similar-looking ocean features. Evaluating multiple detection strategies will ensure adaptability to large-scale maritime monitoring and enable continuous automated operation. The framework will be designed for modularity, allowing future integration of additional sensors or higher-resolution satellite imagery to further boost detection capabilities.

The system will also feature a **drift prediction module** that uses real-time data on ocean currents, wind, tides, and waves to estimate future slick trajectories. This will allow for **faster, targeted containment actions**, source identification, and assessment of potential spread to sensitive habitats. Predictions will be continuously refined as updated environmental data becomes available, ensuring accuracy even in dynamic ocean conditions.

The final outcome will be a **comprehensive real-time monitoring and prediction tool** that improves spill detection accuracy, enables proactive tracking, and supports rapid decision-making for emergency response. By accounting for drift and environmental conditions, it will also enhance impact assessments on marine life and coastal ecosystems, thereby reducing ecological damage and strengthening disaster preparedness.

REFERENCES

[1] Yanzhu Dong, Yongxue Liu, Huize Wang (2025) , "Oil Slick Segmentation Using Deep Learning: A Framework for Real World Applications Based on Real World Imaginery" https://doi.org/10.34133/remotesensing.0613

[2] Mahdikhani, F., & Hassannejad Bibalan, M. (2025). Detection of oil slicks in SAR satellite images using Otsu-Bradley's thresholding method. *Majlesi Journal of Electrical Engineering*, 19(2), 1–9. https://doi.org/10.57647/j.mjee.2025.1902.24

[3] Deliang Xiang, Yu Lu, Dingdong Guan, Guannan Li, Jianda Cheng and Bangjie (2025), "Oil Spill Detection in PolSAR Imagery Using Composite Scattering Power Entropy and Multi-Scale Hybrid Feature Fusion Network" https://doi.org/10.1109/JSTARS.2025.3570697

[4] Xu, J., Cheng, M., Li, B., Chu, L., Dong, H., Yang, Y., Qian, S., Huang, Y., & Yuan, J. (2024). Oil slick identification in marine radar image using HOG, random forest, and PSO. *IEEE Geoscience and Remote Sensing Letters*, 21, 1504305. https://doi.org/10.1109/LGRS.2024.3431043

[5] Saidi, M. Z. P. Z. M., Sha'ban, F. M., Al-Marzouki, M. Y., Othman, S. A. G., Chekima, S. K., Nor, R. A. M., Nawi, M. R. M., & Daut, I. M. (2024). "Performance Measures for Validation of Oil Spill Dispersion Models Based on Satellite and Coastal Data". *Journal of Marine Science and Engineering*, *12*(3), 519. https://doi.org/10.1109/JOE.2021.3099562

[6] Elmi, M. M., Lee, Y. J., & Kim, J. R. (2025). Solution for Modelling the Marine Oil Spill Drift (Article): A Review.*Environments*, *9*(12), 173. https://doi.org/10.3390/environments12040132